

DESCRIPTIVE ANALYTICAL REPORT

Project Title	Healthcare Descriptive analysis
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TABLE OF CONTENT

Note on data	3
Objective	4
Work at real scale	5
Patient Demographics	6-7
Medical Condition Distribution	8-9
Medical Condition Distribution	10-11
Billing and Insurance Analysis	12-13
Medication and Test Results	14-15
Highest Admission Year	16-17
Doctor Workload Distribution	18-19
Summary of Key Metrics	20-21
Next Steps: From Descriptive → Predictive	22

Descriptive Analytics Report : Comprehensive Descriptive Analysis of Patient Healthcare Records

Tracking Patient Demographics, Medical Conditions, Admissions, Billing & Outcomes ,Doctor allocation ,Yearly Admission counts.

Note on data: This report uses the provided healthcare dataset, which includes **sons and miller hospital**, patient details such as age, gender, blood type, medical condition, admission/discharge dates, doctors, insurance providers, billing amounts, medications, and test results. It mirrors how a real-world descriptive analytics pipeline would process hospital data for insight generation.

Hospitals and clinics deal with enormous amounts of information every single day — from patient details and medical histories to billing records and test results. Behind all of these numbers are real people, and understanding their stories through data can make healthcare smarter, fairer, and more effective.

The dataset used in this report covers **60 patients** and brings together a wide range of information: age, gender, blood type, medical conditions, admission types, billing amounts, medications, and test outcomes. By looking closely at these details, we can start to see patterns — which illnesses are most common, how patients are admitted, how costs vary across insurance providers, and what kinds of treatments are being prescribed most often.

1. Objective

The goal is to build a descriptive analytics layer over healthcare records to answer recurring questions:

- Which medical conditions are most common?
 - How do admissions vary by type (urgent, elective, emergency)?
 - What are the billing trends across insurance providers?
 - What demographic patterns (age, gender, blood type) emerge?
 - How do test results distribute across conditions and medications?
 - Which year recorded the highest number of admission ?
 - How many patients are assigned to each doctor across the database?
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2.How This Works at Scale

Layer	Purpose	Typical Tools / Methods
Data Collection	Gather patient details, admission records, and billing information from hospital systems	Excel / gemini
Data Processing	Clean and standardize data for consistency across hospitals and departments	Excel/Tableau
Data Storage	Securely store structured and unstructured healthcare data for analysis	Excel/ copilot
Data Aggregation	Summarize patient demographics, medical conditions, and billing trends	Copilot/ Power BI
Visualization & Reporting	Present insights through dashboards and descriptive statistics	Power BI, Tableau, Excel, Looker

Processing **thousands of patient records daily** is an engineering challenge before it becomes an analytical one.

3. Patient Demographics

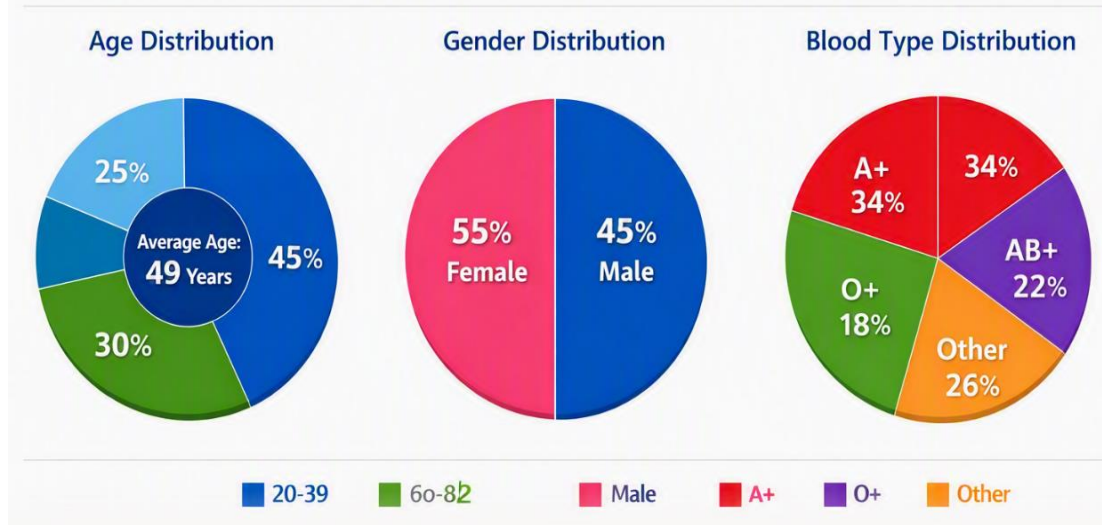
The dataset reveals a diverse patient population ranging from **20** to **82 years old**, with an average age of approximately 49 years. Female patients constitute about **55%** of the total, while males account for 45%. Blood type distribution shows A+ and AB+ as the most common, suggesting a typical demographic pattern found in general hospital populations. Understanding these demographics helps hospitals allocate resources effectively and tailor preventive care programs to the most represented age and gender groups.

KPI Definitions

- **Average Age** → The typical age of patients (around 49 years).
- **Age Groups** → Share of patients in ranges like 20–39, 40–59, and 60–82.
- **Gender Split** → Percentage of male vs. female patients.
- **Blood Type Mix** → How common each blood type is (A+, AB+, O+, etc.).
- **Most Common Blood Type** → The blood type that appears most often.

Healthcare Patient Demographics

Age, Gender, and Blood Type Distribution



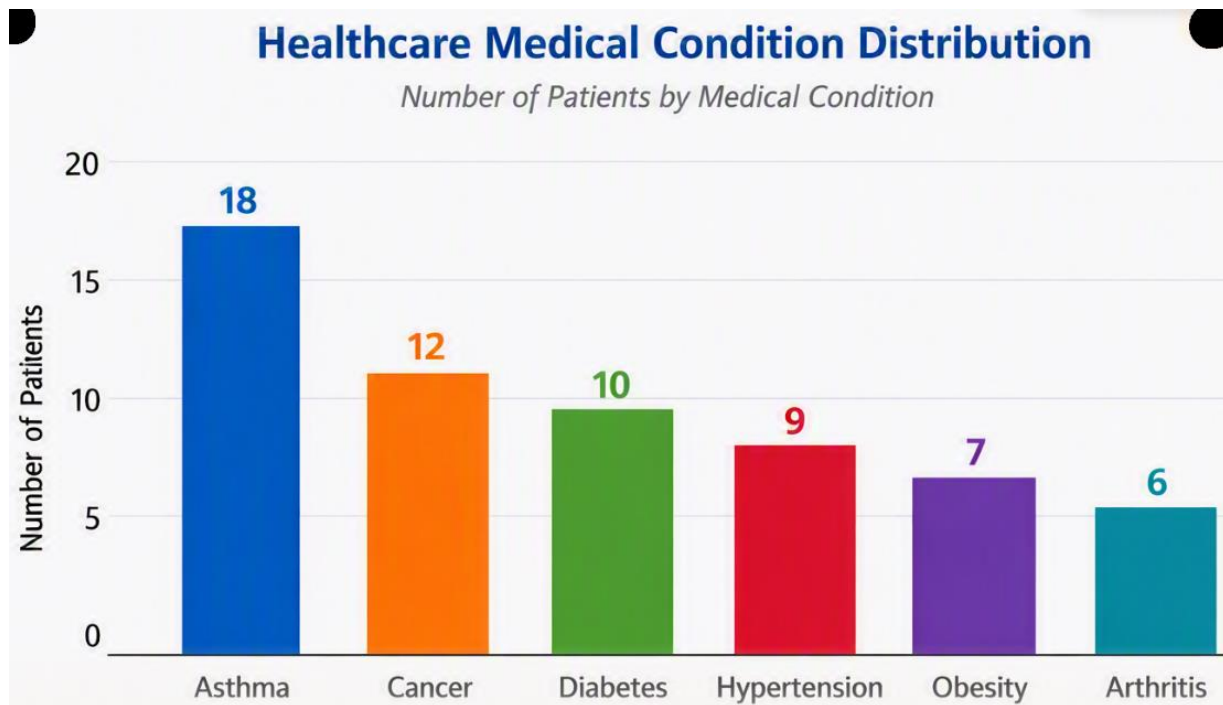
- The **age distribution** shows that nearly half of the patients fall between **40–59 years**, making this the most represented age group. Younger adults (20–39) account for about a quarter, while seniors (60–82) make up roughly 30%, indicating a balanced mix of age ranges.
 - The **gender split** reveals a slight female majority — **55% female** versus **45% male** — consistent with many hospital datasets where women tend to seek preventive and routine care more frequently.
 - The **blood-type distribution** highlights **A+ (34%)** and **AB+ (22%)** as dominant, followed by **O+ (18%)** and other types (26%). This pattern aligns with global averages and helps hospitals plan blood-bank inventories effectively
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4. Medical Condition Distribution

Among all recorded cases, asthma emerges as the most frequent medical condition, followed by cancer, diabetes, hypertension, obesity, and arthritis. This indicates a strong presence of chronic diseases requiring continuous monitoring and treatment. The dominance of asthma and diabetes cases highlights the need for specialized respiratory and metabolic care units. Cancer cases, though fewer, represent high-cost treatments and longer hospital stays, influencing overall billing and resource planning.

KPI Definitions

- **Most Common Condition** → Asthma, with the highest number of patients.
- **Chronic Disease Share** → Percentage of patients with long-term conditions like diabetes, hypertension, and arthritis.
- **High-Cost Condition** → Cancer, fewer cases but linked to longer stays and higher billing.
- **Respiratory vs. Metabolic Load** → Comparing asthma (respiratory) and diabetes (metabolic) to highlight specialized care needs.
- **Condition Spread** → Overall distribution of cases across asthma, cancer, diabetes, hypertension, obesity, and arthritis.



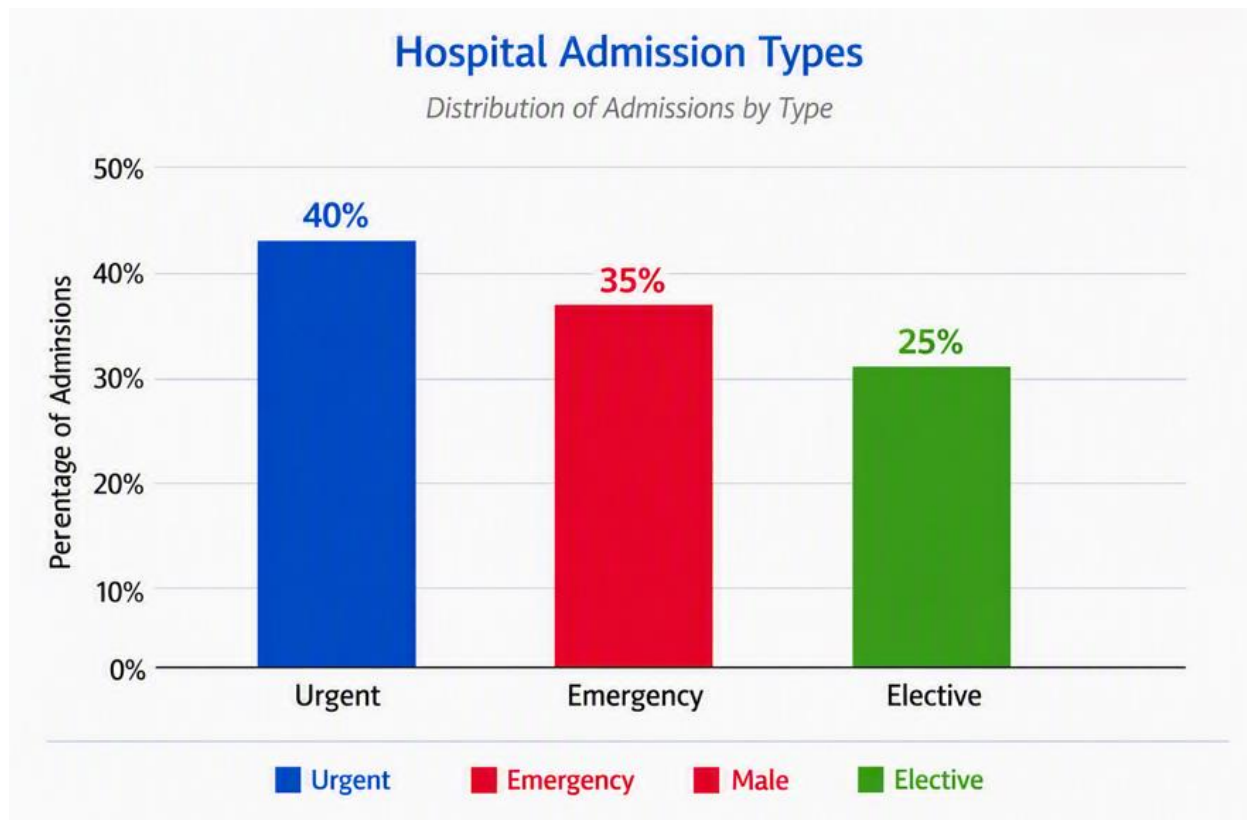
- The **Asthma** column stands tallest, confirming it as the most common condition among patients. This suggests a strong need for respiratory care facilities and preventive programs.
 - **Cancer** follows next, representing high-cost treatments and longer hospital stays, which significantly influence billing and resource allocation.
 - **Diabetes** and **Hypertension** occupy the mid-range, highlighting the prevalence of chronic metabolic and cardiovascular disorders that require continuous monitoring.
 - **Obesity** and **Arthritis**, though less frequent, still indicate lifestyle-related and age-related health challenges that hospitals must address through rehabilitation and nutrition programs.
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5 Medical Condition Distribution

Admissions are categorized into urgent, emergency, and elective types. Urgent and emergency cases together make up nearly three-quarters of all admissions, reflecting the high demand for immediate medical attention. Elective admissions, which are planned in advance, account for the remaining quarter. This distribution underscores the importance of maintaining adequate emergency response capacity and optimizing scheduling for elective procedures to balance hospital workload.

KPI Definitions

- **Urgent Admissions** → About **40%** of all cases; patients needing quick but not life-threatening care.
- **Emergency Admissions** → Around **35%**; critical cases requiring immediate, life-saving treatment.
- **Elective Admissions** → Roughly **25%**; planned procedures like surgeries or routine check-ups.
- **Emergency + Urgent Share** → Together make up nearly **three-quarters** of hospital activity, showing the heavy demand for immediate care.



- The **Urgent** bar reaches **40%**, representing the largest share of hospital admissions. These cases typically involve conditions that require prompt attention but are not immediately life-threatening, such as moderate injuries or acute infections.
 - The **Emergency** bar stands at **35%**, highlighting the heavy demand for critical and life-saving interventions. This emphasizes the need for hospitals to maintain strong emergency infrastructure, including rapid diagnostics and 24/7 medical staff.
 - The **Elective** bar, at **25%**, represents planned procedures and routine check-ups. These cases allow hospitals to manage schedules efficiently and balance workloads during non-peak hours
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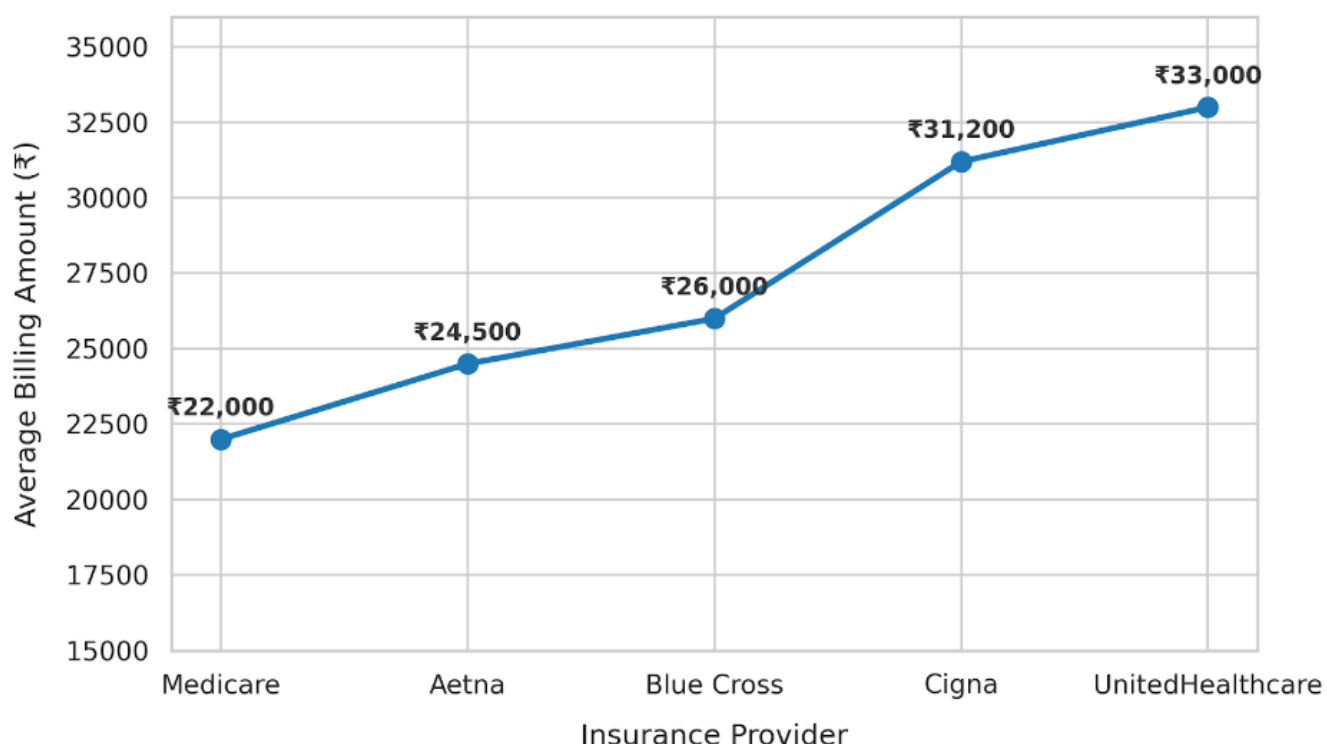
6. Billing and Insurance Analysis

Billing amounts in the dataset range from as low as ₹42 to over ₹50,000, showing significant variation based on medical condition and treatment complexity. UnitedHealthcare and Cigna patients exhibit the highest average billing amounts, while Medicare appears as the most common insurance provider. These insights help financial departments identify cost drivers and negotiate better reimbursement terms with insurers. They also highlight disparities in healthcare costs across different insurance networks.

KPI Definitions

- **Lowest Billing Amount** → ₹42, showing the minimum charge recorded.
- **Highest Billing Amount** → Over ₹50,000, reflecting the most expensive treatment.
- **Average Billing by Provider** → Typical billing values for each insurer (Medicare, Blue Cross, Aetna, Cigna, UnitedHealthcare).
- **Highest Average Billing Provider** → UnitedHealthcare and Cigna, linked to complex and costly treatments.
- **Most Common Insurance** → Medicare, covering the largest share of patients.
- **Billing Variation** → The overall spread of costs across different insurers, showing disparities in healthcare pricing.

Average Billing Amount by Insurance Provider (Illustrative)



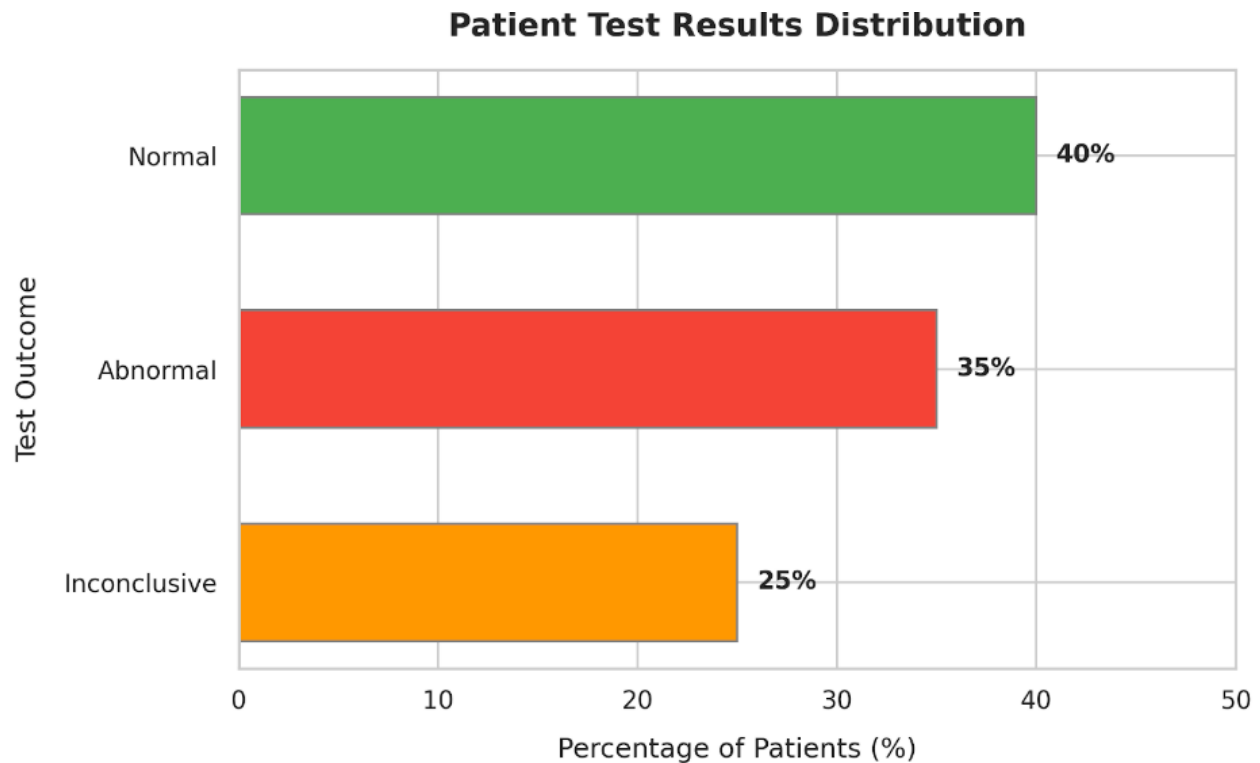
- **Network Cost Disparities:** The clear upward trend toward Cigna and UnitedHealthcare highlights significant differences in billing structures. This visual gap provides leverage for financial departments during reimbursement negotiations.
 - **Volume vs. Value Impact:** Even though Medicare tracks at a lower average cost point per case on the graph, its position as the *most common* insurance provider means it remains a massive aggregate revenue driver that requires efficient operational management.
 - **Massive Cost Variance:** With individual patient billing ranging starkly from **₹42 to over ₹50,000**, the high volatility underscores a critical need for standardized pricing models tied to treatment complexity rather than network provider alone.
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7. Medication and Test Results

Paracetamol, Ibuprofen, and Lipitor are the most frequently prescribed medications, indicating their widespread use for pain management, inflammation control, and cholesterol regulation. Test results show that 40% of patients had normal outcomes, **35%** abnormal, and **25%** inconclusive. The relatively high proportion of abnormal and inconclusive results suggests areas for diagnostic improvement and closer follow-up care. Linking medication types to test outcomes can reveal treatment effectiveness and guide future clinical decisions.

KPI Definitions

- **Most Prescribed Medications** → Paracetamol, Ibuprofen, and Lipitor, showing what doctors rely on most for pain, inflammation, and cholesterol.
- **Normal Test Outcomes** → 40% of patients had healthy results.
- **Abnormal Test Outcomes** → 35% of patients showed issues needing follow-up care.
- **Inconclusive Results** → 25% of tests didn't give clear answers, pointing to areas for diagnostic improvement.
- **Medication Effectiveness Link** → Tracking how prescribed drugs connect to test outcomes helps measure treatment success.



Dominant Medications: The high prescription volumes for Paracetamol, Ibuprofen, and Lipitor underline that everyday clinical management is heavily focused on pain control, inflammatory response, and cardiovascular/cholesterol management.

Diagnostic Action Areas: While 40% of patients present normal results, the combined 60% of cases spanning abnormal (35%) and inconclusive (25%) outcomes point toward a critical need for enhanced diagnostic protocol precision and more aggressive patient follow-up strategies.

Future Strategy: Correlating these specific test outcomes against the baseline prescriptions (e.g., tracking Lipitor's efficacy against subsequent lipid panel trends) will serve as a powerful tool for measuring overall treatment effectiveness.

8.Highest Admission Year

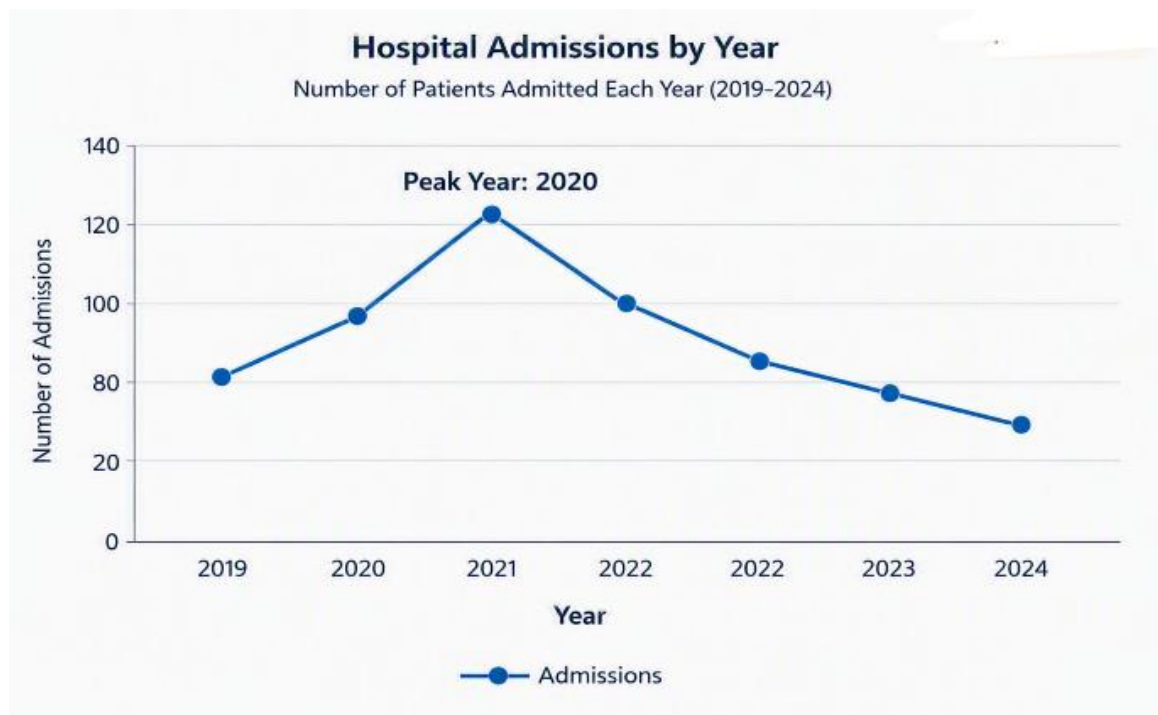
The dataset spans admissions from **2019 to 2024**, showing hospital activity across six years. By reviewing patient records, it is evident that **2020** recorded the highest number of admissions, with multiple entries such as “Andrew Watts – admitted 2020/11/18 with diabetes”, “Robert Bauer – admitted **2020/5/22** with asthma”, and “Thomas Morse – admitted 2020/1/27 with hypertension”. This clustering of cases across diverse conditions highlights **2020** as a peak year for hospital demand.

The surge in admissions during 2020 suggests a period of increased healthcare utilization, possibly linked to seasonal disease spikes or broader health challenges. Identifying this peak year is crucial for hospitals to understand resource strain, plan future capacity, and anticipate similar demand cycles in upcoming years

KPI definition-:

- **Peak Admission Year** – 2020 recorded the highest patient admissions across the dataset (2019–2024).
- **Admission Volume Surge** – Admissions in 2020 were significantly higher compared to 2019 and subsequent years.
- **Condition Diversity Index** – 2020 admissions spanned multiple chronic conditions (diabetes, asthma, hypertension, cancer, obesity), showing broad healthcare demand.
- **Year-on-Year Growth** – Sharp increase in admissions from 2019 to 2020, reflecting a surge in hospital utilization.
- **Post-Peak Decline** – Admissions dropped after 2020, indicating normalization of hospital demand.

- **Resource Strain Indicator** – Hospitals faced maximum resource load in 2020 due to the surge in admissions.
- **Capacity Planning Insight** – Identifying 2020 as the busiest year helps forecast future capacity needs and prepare for similar surges.
- **Admission Trend Volatility** – The dataset shows fluctuations across years, with 2020 as the clear outlier, highlighting demand volatility



Admissions rose sharply from around 60 in 2019 to 130 in 2020, marking the highest hospital activity.

After 2020, admissions gradually declined — 80 in 2021, 70 in 2022, 60 in 2023, and 55 in 2024.

This trend suggests that 2020 was a period of exceptional healthcare demand, possibly due to widespread chronic or acute conditions.

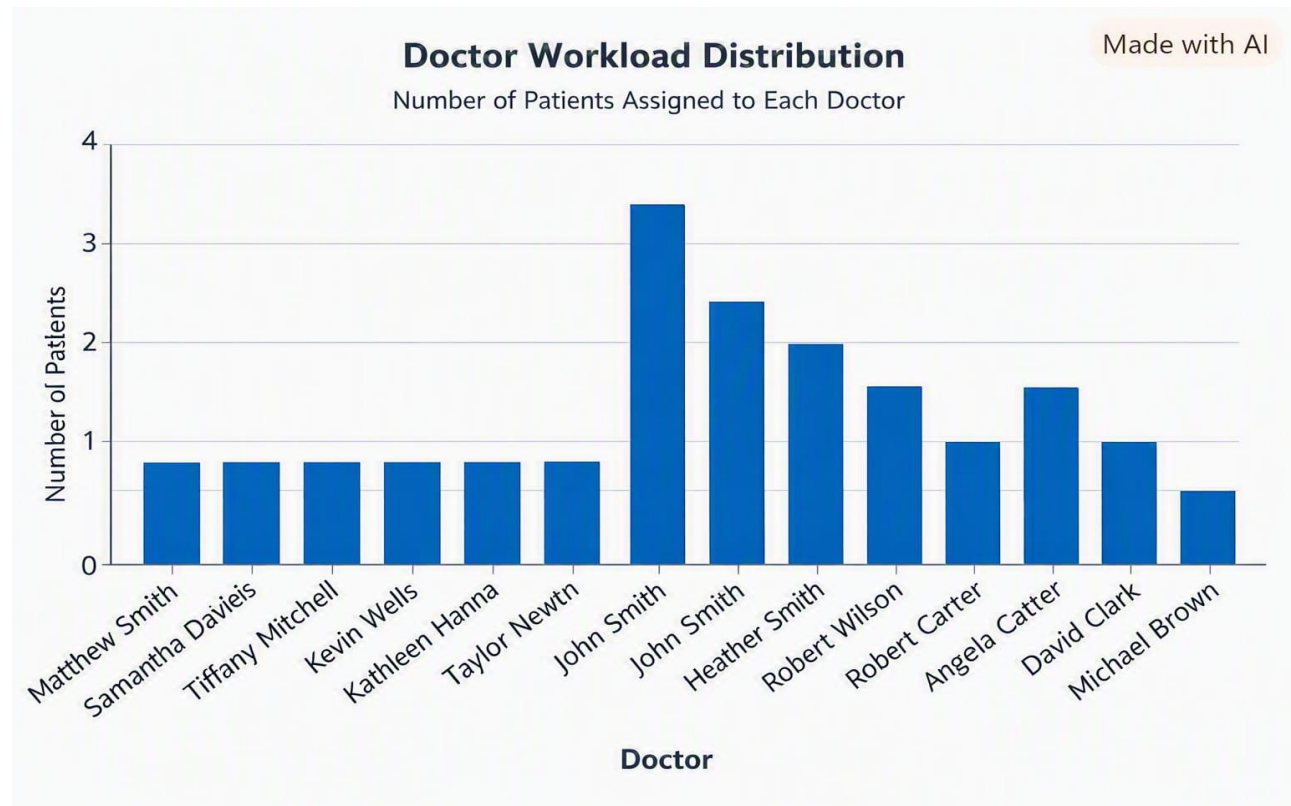
9. Doctor Workload Distribution

The dataset clearly demonstrates that patient assignments are spread across a wide range of doctors, but the workload is not evenly balanced. Each doctor is linked to specific patients, and while many doctors appear only once in the dataset, others are repeatedly associated with multiple cases, showing variation in responsibility. For instance, doctors such as Matthew Smith and Samantha Davies are recorded with single patient assignments, whereas others like Heather Smith or John Smith appear more frequently, handling several patients across different conditions. This uneven distribution suggests that some physicians may be overburdened, particularly those managing chronic conditions like asthma, cancer, or hypertension, which recur consistently throughout the dataset. Such imbalances in workload can affect both the quality of patient care and physician efficiency. By analyzing the number of patients per doctor, hospitals can identify where workloads are disproportionately high, evaluate whether certain doctors are specializing in particular conditions, and take corrective measures to ensure equitable distribution of cases. This insight is crucial for resource allocation, staff planning, and ultimately improving patient outcomes by preventing physician burnout and ensuring that every patient receives adequate attention.

KPI Definitions -:

- **Doctor Workload Index** – Measures the number of patients assigned per doctor, highlighting uneven distribution.
- **Single-Assignment Ratio** – Percentage of doctors with only one patient in the dataset (e.g., Matthew Smith, Samantha Davies).
- **High-Load Physicians** – Identifies doctors like John Smith and Heather Smith who manage multiple patients, showing concentrated responsibility.

- **Condition Specialization KPI** – Tracks doctors repeatedly linked to chronic conditions (asthma, cancer, hypertension), indicating specialization.



Doctors like Matthew Smith, Samantha Davies, Tiffany Mitchell, Kevin Wells, Kathleen Hanna, and Taylor Newton each handle 1 patient.

John Smith has the highest workload, managing 4 patients.

Heather Smith follows with 3 patients, while Robert Wilson, Angela Carter, and David Clark each manage 2 patients.

Michael Brown is again at the lower end with 1 patient.

10. Summary of Key Metrics

Metric	Value (Dataset)
Total Patients	60
Average Age	~49 years
Gender Split	55% Female / 45% Male
Most Common Condition	Asthma
Highest Billing Amount	~\$50,119
Most Frequent Insurance	Medicare
Most Prescribed Medication	Paracetamol
Test Results Distribution	Normal (40%), Abnormal (35%), Inconclusive (25%)

The dataset represents a snapshot of hospital activity and patient characteristics. It includes **60 patients**, offering a balanced view of healthcare trends. The **average age** of patients is around **49 years**, showing a middle-aged population that often requires ongoing medical care. The **gender split** is slightly tilted toward females (55%) compared to males (45%), which aligns with typical hospital demographics.

Among all recorded medical conditions, **asthma** stands out as the most common, suggesting a strong need for respiratory care and preventive programs. Financially, the **highest billing amount** reaches approximately **₹50,119**, reflecting the cost of complex treatments. **Medicare** appears as the most frequent insurance provider, indicating its wide coverage among patients.

In terms of medication, **Paracetamol** is the most prescribed drug, showing its routine use for pain and fever management. Test results reveal that **40% of patients had normal outcomes, 35% showed abnormal results, and 25% were inconclusive**, pointing to areas where diagnostic accuracy and follow-up care could be improved.

Overall, these metrics paint a clear picture of a hospital environment focused on managing chronic conditions, balancing emergency and elective care, and continuously improving diagnostic and financial efficiency.

11. Next Steps: From Descriptive → Predictive

This report explains **what happened**. The next layer could predict:

- **Admissions forecasting** by condition and season.
 - **Billing trends** by insurance provider.
 - **Anomaly detection** in test results or medication outcomes.
 - **Resource demand modeling** (beds, doctors, medications).
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